Machine Learning: Practice Midterm

Name: _____

1. Evaluating classifiers (20 points)

A charity trains African giant pouched rats ("HeroRATs") to detect tuberculosis (TB) via their impressive sense of smell. To make sure the rats are effective, each rat is tested on 1000 samples of sputum. Suppose a laboratory test shows that 100 of them come from patients with TB. The rat pauses at 80% of the TB samples and at 10% of the others. Treat "has TB" as the positive class.

1.1 [4 points] Write down the confusion matrix.

1.2 [2 points] What are precision and recall for this rat?

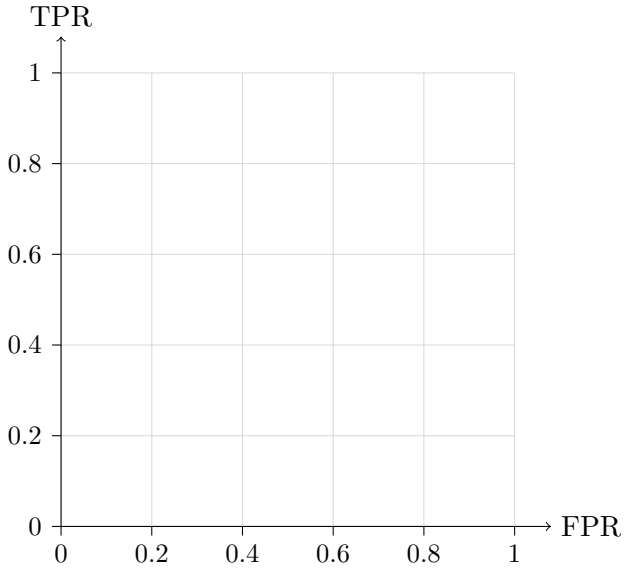
1.3 [2 points] Do we care more about improving precision or recall for this problem, and why?

The rat handler also records how long the rat pauses at each sample before moving on. Here are 10 samples, 5 from TB patients and 5 without TB:

sample	A	B	C	D	E	F	G	H	I	J
pause (seconds)	2.6	0.6	4.6	2.0	1.3	3.7	4.2	1.0	3.0	2.4
TB?	yes	no	yes	no	yes	no	yes	no	yes	no

We say a rat flags a sample as TB if it pauses for over t seconds.

1.4 [10 points] Draw the ROC curve.



1.5 [2 points] Compute the AUC.

2. Support vector machines (20 points)

2.1 [10 points] A soft-margin SVM returned $\boldsymbol{w} = (4, 3)$, $b = -12$.

point	A	B	C	D	E
x_1	3	1	2.5	1	3
x_2	1	2	1	2.5	0.5
label	+	−	+	−	−

(a) Compute $f(\boldsymbol{x})$ and the predicted class of each point.

(b) What is the width of the margin?

(c) Compute each point's distance to the decision boundary.

(d) Which points are support vectors? Which point is on the margin, which is inside it, and which is misclassified?

2.2 [6 points] Write down the objective function for the soft-margin SVM and explain why this is what we want to optimize.

2.3 [2 points] Suppose our soft-margin SVM obtains 95% training accuracy but 60% validation accuracy. What does this suggest about our hyperparameter C?

2.4 [2 points] What are support vectors? Is every training datapoint that falls inside the margin a support vector?

3. Nonlinear regression (20 points)

3.1 [4 points] A regression tree is trained on six points:

x	1	2	3	4	5	6
y	1	3	2	7	9	8

Consider the split $x < 3.5$. What does each side predict? What is the SSE of this split?

3.2 [6 points] What happens to the fit, toward overfitting or toward underfitting, as you increase (a) the maximum depth of a regression tree, (b) k in kNN regression, (c) ε in SVR? Explain each.

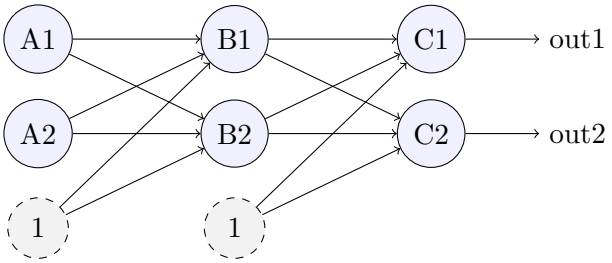
3.3 [5 points] In a training set the targets y range from 10 to 50. Can (a) a regression tree, (b) kNN regression, (c) SVR predict $y = 60$ for some new input? Explain.

3.4 [5 points] A model predicts crop yield from rainfall (in mm, 0 to 2,000) and average temperature (in °C, 10 to 35). We change rainfall to meters, so it now ranges from 0 to 2. For each of a regression tree, kNN and SVR, do the predictions change? Why or why not?

4. Neural networks (30 points)

In this question, row j of a weight matrix $\boldsymbol{W}^{(l)}$ holds the weights into node j of layer l: $W^{(l)}_{ji}$ is the weight from node i of the previous layer to node j. The bias of node j is $b^{(l)}_j$, the weight on the input that is always 1.

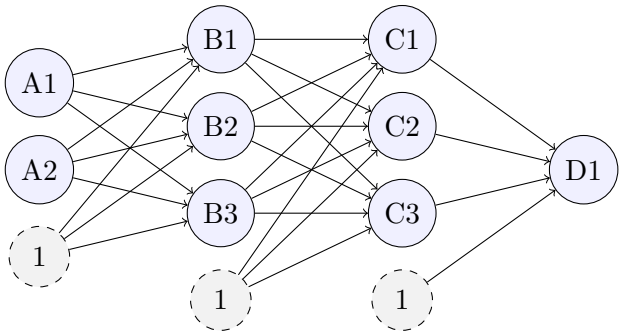
4.1 [8 points] In the network below, out1 represents Class A and out2 represents Class B. All nodes use the sign function as their activation function: $\text{sign}(z) = +1$ if $z > 0$ and -1 if $z < 0$.



$$\boldsymbol{W}^{(1)} = \begin{pmatrix} 0.5 & -0.3 \\ -0.4 & 0.6 \end{pmatrix}, \quad \boldsymbol{b}^{(1)} = \begin{pmatrix} 0.2 \\ -0.5 \end{pmatrix}, \quad \boldsymbol{W}^{(2)} = \begin{pmatrix} 0.7 & 0.4 \\ -0.6 & 0.3 \end{pmatrix}, \quad \boldsymbol{b}^{(2)} = \begin{pmatrix} 0.2 \\ 0.1 \end{pmatrix}$$

Show the forward pass, in matrix form, of the record with Feature1 = 6 and Feature2 = 4. How would this network classify the record (Class A or Class B)?

4.2 [14 points] The network below has three layers of weights. A record with input $A_1 = 2$, $A_2 = 1$ and an actual value $Y = 4$ gets passed in.



	function	derivative
B layer: sigmoid	$f(z) = 1/(1 + e^{-z})$	$f(z)\,(1 - f(z))$
C layer: ReLU	$f(z) = 0$ if $z < 0$; $f(z) = z$ if $z \geq 0$	0 if $z < 0$; 1 if $z \geq 0$
D layer: linear	$f(z) = z$	1
loss: squared error	$E = \frac{1}{2}(Y' - Y)^2$	$\partial E/\partial Y' = Y' - Y$

Here Y' is the network's output, D1. The weights during the forward propagation are

$$\boldsymbol{W}^{(1)} = \begin{pmatrix} 0.5 & -1.0 \\ 1.0 & 0.5 \\ -0.5 & 1.5 \end{pmatrix}, \boldsymbol{b}^{(1)} = \begin{pmatrix} 0.5 \\ -1.0 \\ 0 \end{pmatrix}, \quad \boldsymbol{W}^{(2)} = \begin{pmatrix} -1.0 & 0.5 & -1.0 \\ 2.0 & 1.0 & -0.5 \\ 0.5 & 1.5 & 1.0 \end{pmatrix}, \boldsymbol{b}^{(2)} = \begin{pmatrix} 0.5 \\ -0.5 \\ -1.0 \end{pmatrix},$$

$$\boldsymbol{W}^{(3)} = \begin{pmatrix} 1.5 & -0.4 & 2.0 \end{pmatrix}, \quad b^{(3)} = 0.5.$$

You do not need to forward propagate. The node output values during the forward propagation are

A layer	B layer	C layer	D layer
$A_1 = 2$	$B_1 = 0.62$	$C_1 = 0$	$D_1 = 2.32$
$A_2 = 1$	$B_2 = 0.82$	$C_2 = 1.25$	
	$B_3 = 0.62$	$C_3 = 1.16$	

Adjust $W_{21}^{(1)}$, the weight from A1 to B2, using a learning rate of 1.

4.3 Short answers (8 points in all).

(a) [3 points] Suppose the output D1 used a sigmoid, the label were $Y \in \{0, 1\}$, and the loss were cross-entropy (log loss) instead of squared error. Which step of your backpropagation in 4.2 would change, and how?

(b) [3 points] Why do the hidden layers need nonlinear activation functions? What would the network in 4.2 compute if every activation were linear?

(c) [2 points] While training a network, the training loss keeps going down, but the validation loss is lowest at epoch 40 and then rises. What is happening, and what would you do?